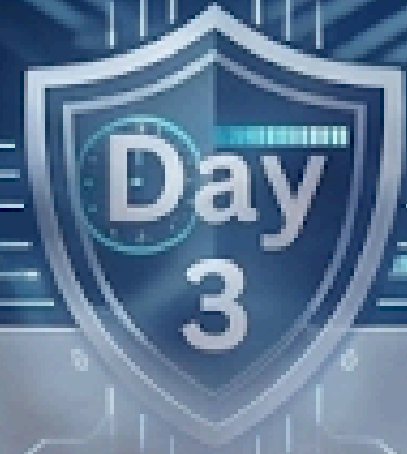




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Course: Diploma in Cyber Security

DAY 3 – RAM AND STORAGE

Topic: RAM and Storage



WHAT IS RAM?

RAM – Random Access Memory

RAM is the computer's temporary working memory. It keeps the data and programs that are currently being used so that the CPU can access them quickly.

RAM is called temporary memory because its contents are normally lost when the computer is switched off.

Simple example

If I open Chrome, Spotify and a game at the same time, these running programs need RAM to keep their active data ready.

RAM = Current work

DDR3, DDR4 AND DDR5

What does DDR mean?

DDR stands for Double Data Rate. DDR is a type of computer memory technology used in RAM. Different DDR generations have different speeds, voltage requirements and compatibility.

Feature	DDR3	DDR4	DDR5
Typical data rates	About 800–2133 MT/s	About 1600–3200 MT/s	About 3200–6400+ MT/s
Standard voltage	1.5 V	1.2 V	1.1 V
Generation	Older	Common/previous generation	Newer generation
Typical use	Older PCs/laptops	Many PCs from recent years	Newer PCs/laptops
Compatibility	DDR3 slot	DDR4 slot	DDR5 slot

Important point

DDR3, DDR4 and DDR5 are not interchangeable.

A DDR4 RAM stick cannot normally be inserted into a DDR5 motherboard because the physical design and electrical requirements are different.

DDR3 → Older

DDR4 → Middle generation

DDR5 → Newer + faster

HOW MUCH RAM DO WE NEED?

RAM and everyday tasks

The amount of RAM needed depends on what we are doing.

Basic work

For browsing, documents and simple applications, a moderate amount of RAM is usually enough.

Gaming

Games generally need more RAM than basic office work, especially when other applications are running at the same time.

Multitasking

If I keep many browser tabs, Spotify, a coding application and other programs open together, more RAM gives the system more working space.

Important idea

More RAM does not automatically make every computer faster.
If the computer already has enough RAM for the task, adding much more may not produce a major performance improvement.

RAM capacity = How much active work the PC can keep ready

WHAT IS COMPUTER STORAGE?

Storage

Storage is where a computer keeps data for long-term use.

Unlike RAM, storage keeps data even after the computer is switched off.

It can contain:

- Operating system
- Applications
- Photos
- Videos
- Documents
- Games
- Personal files

Three storage types for this topic:

HDD → Mechanical storage

SSD → Flash storage

NVMe SSD → High-speed SSD using the NVMe protocol over PCIe

Easy comparison

RAM = Temporary

Storage = Permanent

Simple example

When I download a game, the game files are saved on my SSD/HDD. When I start the game, some of its required data is loaded from storage into RAM.

HDD VS SSD VS NVMe

Feature	HDD	SATA SSD	<u>NVMe</u> SSD
Technology	Magnetic disk	Flash memory	Flash memory
Moving parts	Yes	No	No
Speed	Slowest of the three	Faster than HDD	Usually fastest
Connection	Usually SATA	Usually SATA	Usually PCIe
Best for	Low-cost large storage	Everyday fast storage	Fast applications and demanding workloads
Example use	Backup/archive	General PC use	Gaming, editing, development

1. HDD – Hard Disk Drive

An HDD stores data on spinning magnetic disks. It is usually cheaper for large storage capacities, but it is slower and has moving mechanical parts.

Mechanical

2. SSD – Solid State Drive

An SSD stores data using flash memory and has no spinning disks. It is faster and more resistant to physical shock than an HDD.

Flash

3. NVMe SSD

NVMe is a storage protocol designed for fast communication with SSDs over PCIe. NVMe SSDs can provide much higher performance and lower latency than traditional SATA-based storage.

High-speed SSD

WHICH STORAGE IS USEFUL FOR WHICH APP?

1. HDD – Backup and large files

Useful for:

Movies, photos, backups and large files

Why?

HDDs can provide a lot of storage at a relatively low cost.

2. SATA SSD – Everyday applications

Useful for:

Windows, Chrome, Office, Spotify and normal applications

Why?

An SSD makes booting and application loading much faster than an HDD.

3. NVMe SSD – Gaming and demanding work

Useful for:

Modern games, video editing, programming projects and large file transfers

Why?

NVMe SSDs can provide very high storage performance and low latency.

Easy memory trick:

HDD = Cheap large storage

SSD = Fast everyday storage

NVMe = Very fast storage

PRACTICAL: RAM REMOVAL AND REINSTALLATION

Before touching RAM

Always shut down the computer, disconnect power, and follow the device manufacturer's instructions.

For a laptop, the internal battery may also need to be disconnected depending on the model.

Basic process

Step 1: Shut down the computer.

Step 2: Disconnect the power.

Step 3: Open the case/back cover only if you are allowed and know how to do it safely.

Step 4: Locate the RAM slot.

Step 5: Release the retaining clips.

Step 6: Carefully remove the RAM module by holding its edges.

Step 7: Check the notch and align the RAM correctly with the slot.

Step 8: Press the module into place until the retaining clips lock.

Step 9: Close the computer and reconnect power.

Step 10: Start the computer and check whether the system recognizes the RAM.

Important safety points

- Do not force the RAM.
- Hold the RAM by its edges.
- Do not touch the gold contacts unnecessarily.
- Make sure the computer is powered off before opening it.
- Follow the manufacturer's manual.



Figure 1: RAM module installed in the computer

PRACTICAL: STORAGE DRIVE

Identifying the storage connection

Before removing a storage device, first identify what type of drive and connector the system uses.

SATA drive

A SATA HDD or SATA SSD normally uses:

- SATA data connection
- Power connection from the PSU

M.2 NVMe SSD

An M.2 NVMe SSD is installed directly into a compatible M.2 slot on the motherboard.

Basic installation idea

Power off → Disconnect → Locate drive → Remove carefully → Reinstall → Reconnect → Boot → Verify

Important safety point

Only open a computer that you are allowed to open.

If you do not have access to a suitable device, do not open someone else's laptop or PC just for the assignment.

Instead, you can use a trusted demonstration video for your specific device model.



Figure 2: Storage drive and its connection

RAM VS STORAGE

The easiest way to understand the difference

RAM	Storage
Temporary working memory	Long-term data storage
Used by active programs	Stores files and applications
Data normally disappears after shutdown	Data remains after shutdown
Helps with multitasking	Determines storage capacity and loading performance
Example: DDR4 RAM	Example: SSD/ <u>NVMe</u> SSD

Real-life example

Think about a study table:

RAM = Your table

It is the space where you keep the books you are currently studying.

Storage = Your cupboard

It keeps all your books even when you are not studying them.

CONCLUSION

(DAY 3 – RAM & STORAGE)

In this practical, I learned about the two most important memory systems of a computer: RAM and Storage. RAM is a temporary memory that helps the computer run active programs smoothly, while storage is a permanent memory that keeps data even when the system is turned off.

I also studied different generations of RAM such as DDR3, DDR4 and DDR5, and understood that each generation has different speed, voltage and compatibility. Similarly, I learned about different storage types like HDD, SSD and NVMe SSD, where HDD is slower but cheaper, SSD is faster, and NVMe SSD provides the highest performance.

Overall, this practical helped me understand how computer performance depends on both RAM and storage working together. RAM handles current tasks, while storage keeps all data safe for long-term use.